SPEEDOMETER DRIVE ADAPTER (Continued)

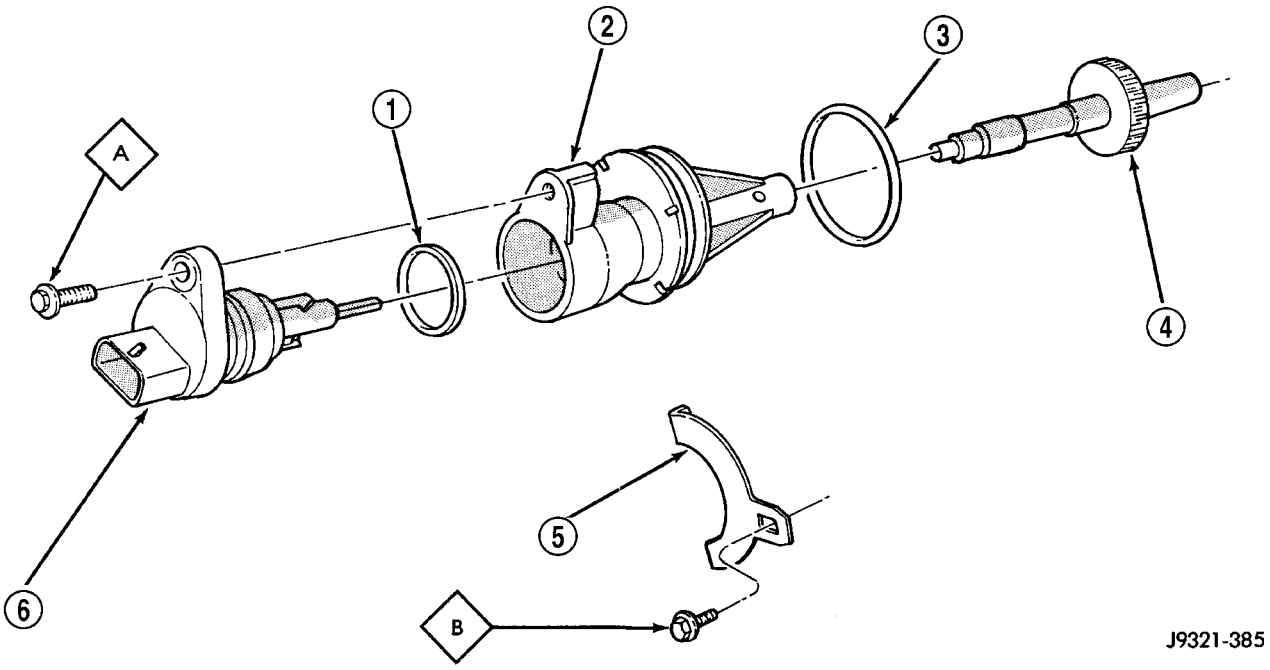

Fig. 98 Speedometer

1 - Sensor O-ring
2 - Speedometer Adapter
3 - Adapter O-ring
4 - Speedometer Pinion
5 - Adapter Clamp
6 - Vehicle Speed Sensor
A - 2-3 N·m (15-27 in. lbs.)
B - 10-12 N·m (90-110 in. lbs.)

(11) Connect wires to vehicle speed sensor.

(12) Lower vehicle and top off transmission fluid level if necessary.

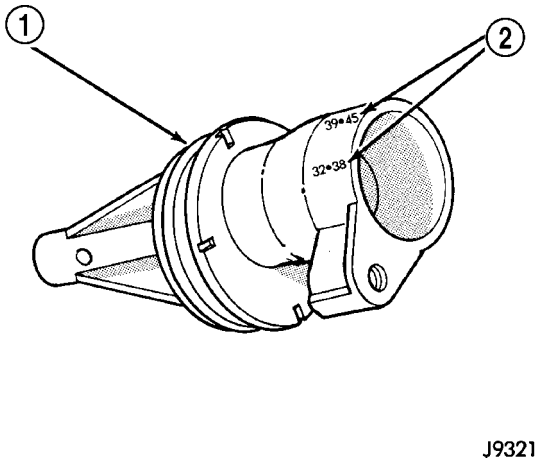

Fig. 99 Location Of Index Numbers On Speedometer Adapter

1 - SPEEDOMETER ADAPTER
2 - INDEX NUMBER LOCATION

VEHICLE SPEED SENSOR

DESCRIPTION

The 3-wire Vehicle Speed Sensor (VSS) is located on the speedometer pinion gear adapter. If equipped with 4WD, this adapter is located on the extension housing of the transfer case (drivers side). If equipped with 2WD, this adapter is located on the left side of the transmission extension housing.

OPERATION

The VSS is a 3-circuit (3–wire), magnetic, hall-effect sensor.

The 3 circuits are:

- A 5-volt power supply from the Powertrain Control Module (PCM).
- A ground is provided for the sensor though a low-noise sensor return circuit in the PCM.
- An input to the PCM is used to determine vehicle speed and distance traveled.

The speed sensor generates 8 pulses per sensor revolution. These signals, in conjunction with a closed throttle signal from the throttle position sensor, indicate a closed throttle deceleration to the PCM. When the vehicle is stopped at idle, a closed throttle signal is received by the PCM (but a speed sensor signal is not received).

Under deceleration conditions, the PCM adjusts the Idle Air Control (IAC) motor to maintain a desired MAP value. Under idle conditions, the PCM adjusts the IAC motor to maintain a desired engine speed.